

Tutorial No : 3

Tutorial Title: Write a program in C++ to sort the numbers in an array using separate functions for read, display, sort and swap. The objective of this assignment is to learn the concepts of input, output, functions, call by reference in C++..

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PROGRAM:

```
#include <iostream>

using namespace std;

void read(int arr[], int n) {

    cout << "Enter " << n << " elements:\n";

    for (int i = 0; i < n; i++)

        cin >> arr[i];

}

void display(int arr[], int n) {

    for (int i = 0; i < n; i++)

        cout << arr[i] << " ";

    cout << endl;

}

void swap(int &a, int &b) {

    int temp = a;

    a = b;

    b = temp;
```

```

}
void sort(int arr[], int n) {

    for (int i = 0; i < n - 1; i++) {

        for (int j = 0; j < n - i - 1; j++) {

            if (arr[j] > arr[j + 1]) {

                swap(arr[j], arr[j + 1]);

            }

        }

    }

}

int main() {

    int n;

    cout << "Enter the size of the array: ";

    cin >> n;

    int arr[n];

    read(arr, n);

    cout << "Original Array: ";

    display(arr, n);

    sort(arr, n);

```

```
cout << "Sorted Array: ";  
  
display(arr, n);  
  
return 0;  
  
}
```

Output

```
Enter the size of the array: 5  
Enter 5 elements:  
1 4 3 5 2  
Original Array: 1 4 3 5 2  
Sorted Array: 1 2 3 4 5  
  
=== Code Execution Successful ===
```

OUTPUT: